

Engineering Mathematics: Matrix Reduction to Normal Form

1. Theoretical Foundation

1.1 The Concept of Normal Form

The **Normal Form** (also known as the Canonical Form) of a matrix A of rank r is a specific block matrix structure obtained using elementary row and column transformations. It is typically represented as:

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

Where I_r is an identity matrix of order r , and the surrounding zeros represent null matrices of appropriate dimensions.

1.2 Objective of the Transformation

The goal is to isolate a leading identity block. Unlike the Echelon form (which only uses row operations), the Normal Form utilizes **both row and column operations** to systematically "clear" all elements except for the diagonal ones.

2. The Logic: Systematic Reduction Algorithm

To transform a matrix into its normal form, follow this hierarchical logic:

- Pivot Selection:** Ensure the element at a_{11} is 1. If it is not, swap rows or columns to bring a 1 to the a_{11} position.
- Row Clearing:** Use the pivot at a_{11} to make all elements in the first column below it ($a_{21}, a_{31}, \dots$) equal to zero using row operations.
- Column Clearing:** Use the pivot at a_{11} to make all elements in the first row to its right ($a_{12}, a_{13}, \dots$) equal to zero using column operations.
- Recurse:** Move to the next diagonal element a_{22} and repeat the process for the remaining sub-matrix.

3. Step-by-Step Solution

Problem: Find the normal form and rank of the matrix:

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

Step 1: Establish a Leading Pivot

We swap R_1 and R_2 to get 1 at the $(1, 1)$ position.

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} (R_1 \leftrightarrow R_2) \longrightarrow \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

Step 2: Clear the First Column

Apply row operations: $R_2 \rightarrow R_2 - 2R_1$, $R_3 \rightarrow R_3 - 3R_1$, $R_4 \rightarrow R_4 - 6R_1$.

$$\begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} (R_2 - 2R_1, R_3 - 3R_1, R_4 - 6R_1) \longrightarrow \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

Step 3: Clear the First Row

Apply column operations: $C_2 \rightarrow C_2 + C_1$, $C_3 \rightarrow C_3 + 2C_1$, $C_4 \rightarrow C_4 + 4C_1$.

$$\begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix} (C_2 + C_1, C_3 + 2C_1, C_4 + 4C_1) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

Step 4: Normalize the Second Pivot

Perform $R_2 \rightarrow R_2 - R_3$ to create a 1 at a_{22} .

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 5 & 3 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix} (R_2 - R_3) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -6 & -3 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$$

Step 5: Clear Column 2 and Row 2

Apply $R_3 \rightarrow R_3 - 4R_2$, $R_4 \rightarrow R_4 - 9R_2$ then $C_3 \rightarrow C_3 + 6C_2$, $C_4 \rightarrow C_4 + 3C_2$.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -6 & -3 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 66 & 44 \end{bmatrix} (C_3 + 6C_2, C_4 + 3C_2) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 66 & 44 \end{bmatrix}$$

Step 6: Final Reduction

Divide R_3 by 33, then clear R_4 and C_4 .

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 33 & 22 \\ 0 & 0 & 66 & 44 \end{bmatrix} (R_3/33, R_4 - 2R_3) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2/3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \left(C_4 - \frac{2}{3}C_3 \right) \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

4. Final Result

The matrix has been reduced to the form:

$$\begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$$

Rank of Matrix A: The number of non-zero rows (or the order of the identity block) is **3**.

$$\rho(A) = 3$$